

# Madhukanth M

## JAVA DEVELOPER | AI & DATA SCIENCE SPECIALIST

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### PROFESSIONAL SUMMARY

Highly motivated B.Tech Artificial Intelligence and Data Science undergraduate and Java Developer with a strong background in data structures, algorithms, and modular software engineering. Experienced in designing low-latency network protocols, automated IoT sensing layers, and containerized computer-vision ingestion pipelines. Proven competitor with awards in national coding challenges and responsive layout sprints. Competent in writing clean, structured, and performant backend code ready to support business critical software divisions.

### TECHNICAL SKILLS

|                        |                                                                                |
|------------------------|--------------------------------------------------------------------------------|
| Languages              | Java, Python, C, C++                                                           |
| Core CS & AI           | Data Structures & Algorithms, Machine Learning Model Training, Computer Vision |
| Tools & Infrastructure | Git, GitHub, Docker, Nginx, Linux OS, IoT Systems, UI/UX Prototyping           |
| Protocols & APIs       | UDP Sockets, TCP/IP, REST APIs, JSON data serialization                        |

### EDUCATION

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| Bannari Amman Institute of Technology                                                                                                                                                                                                                                                                            | Aug 2024 - May 2028 (Expected) |
| Bachelor of Technology (B.Tech) - Artificial Intelligence & Data Science                                                                                                                                                                                                                                         |                                |
| <ul style="list-style-type: none"><li>Core focus in neural networks, statistical data analysis, machine learning model validation, and computational complexity.</li><li>Active technical volunteer for university Code Circle Club, assisting in compiling platforms and routing technical execution.</li></ul> |                                |
| Gandhi Matriculation Higher Secondary School                                                                                                                                                                                                                                                                     | Apr 2022 - Apr 2024            |
| Higher Secondary Certificate (HSC)                                                                                                                                                                                                                                                                               |                                |
| <ul style="list-style-type: none"><li>Acquired foundational expertise in advanced algebra, physics, chemical structures, and computer science principles.</li></ul>                                                                                                                                              |                                |

### FEATURED PROJECTS

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |           |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| v-mark (Wildfire Detection System)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Completed |
| Independent Developer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |           |
| <ul style="list-style-type: none"><li>Engineered a real-time wildfire detection and hazard intelligence system leveraging multi-stage computer-vision models.</li><li>Designed and implemented a scalable Python backend and ingestion REST APIs for low-latency hotspot filtering.</li><li>Integrated drone telemetry data pipelines and field interfaces, lowering data transmission lag.</li><li>Containerized pipeline microservices using Docker and set up Nginx-based frontend serving for deployment ease.</li></ul> |           |
| Source Code                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |           |
| Agri Care (Smart AI Agriculture Platform)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Completed |
| Lead Developer - Bannari Amman Institute of Technology                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |           |
| <ul style="list-style-type: none"><li>Developed an automated crop advisory system polling soil moisture, temperature, and environmental inputs.</li><li>Implemented early crop pest and disease detection routines using custom ML classification models.</li><li>Configured multi-lingual routing and real-time agricultural crop market price ingestion pipelines.</li><li>Boosted farm water conservation and precision irrigation diagnostics through soil telemetry analysis.</li></ul>                                 |           |
| Source Code                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |           |

**Aura Sink (Real-Time Audio Sync Engine)**

Ongoing

**Independent Developer**

- Programmed a low-latency network audio broadcast engine transmitting sync streams to multiple client targets.
- Wrote device clock synchronization protocols and playback drift corrections utilizing UDP sockets.
- Structured custom data ring buffers to absorb network jitter, bringing stream latency down to sub-millisecond offset.

[Source Code](#)**ACHIEVEMENTS & CERTIFICATIONS**

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- **Winner - Code With Curious 2.0:** Placed 1st in national coding and algorithmic problem-solving sprint.
- **Runner-up (Top 2) - FRONT END Frenzy 2.0:** Secured 2nd place in rapid styling and UI design challenge.
- **Oracle Cloud Infrastructure 2025 Certified Professional:** Validated cloud virtualization, VPC configuration, and compute provisioning capabilities.
- **Python Essentials 1 & 2:** Certified by Cisco Networking Academy in algorithmic paradigms, OOP designs, and package architectures.
- **C++ Essentials 1:** Certified in C++ pointer arithmetic, memory indexing structures, and system header definitions.
- **Introduction to Modern AI, Data Science, Cybersecurity, & IoT:** Certified by Cisco Academy in data clearing, IoT serial routing, and network threat defense topologies.
- **IEEE Database Training:** Certified by IEEE Xplore / EBSCO India in literature indexing and database querying.

**VOLUNTEERING & LEADERSHIP**

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**Code Circle Club - Technical Event Volunteer (Celestra)**

2025 - Aug 2025

**Bannari Amman Institute of Technology**

- Supervised participant coding platforms, managed compiler setups, and debugged environmental compilation errors for coding contestants.